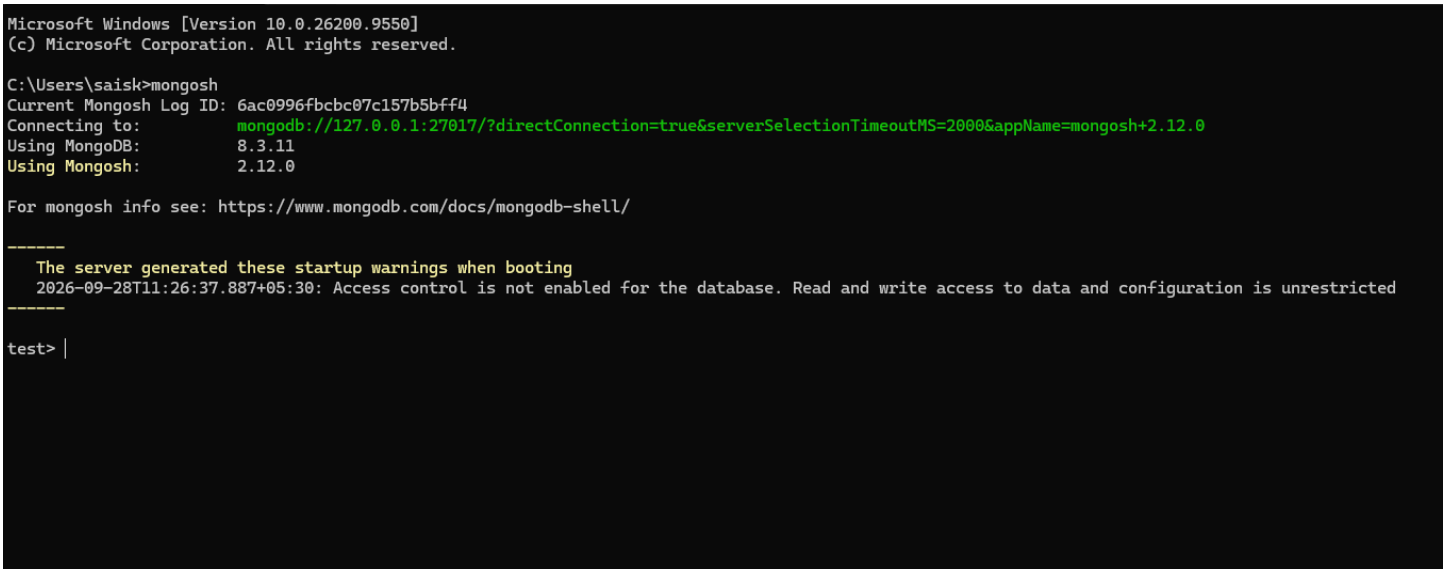
1. **mongosh** — Used this command to connect to my local MongoDB instance and run MongoDB commands through the terminal.



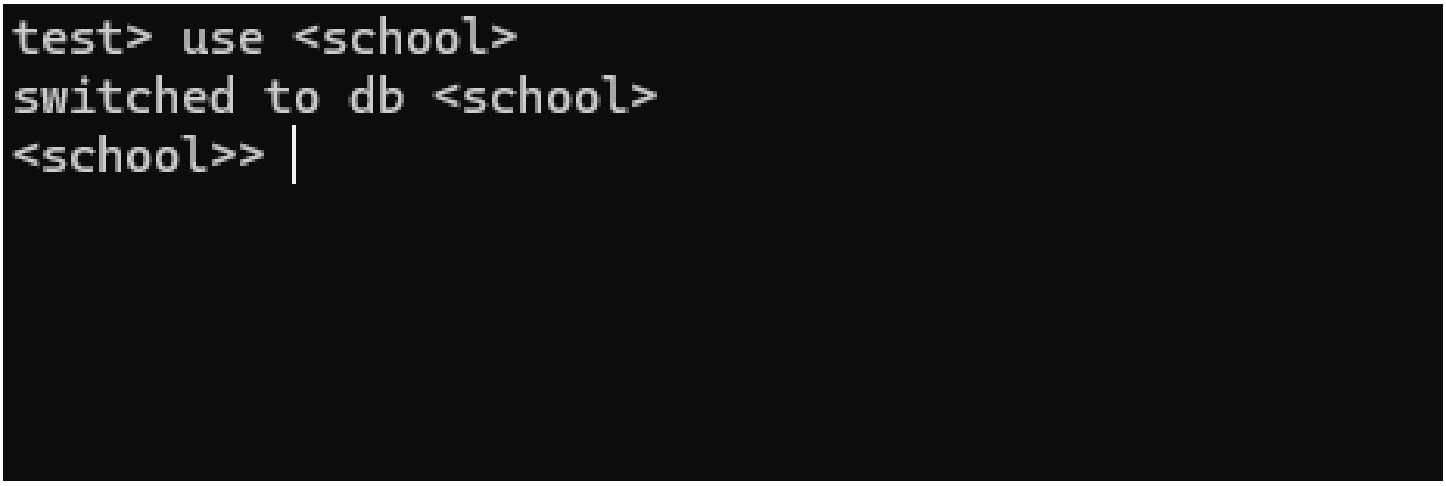
2. **show dbs** — Used this command to display all the databases available in my current MongoDB instance.



3. **db** — Used this command to display the name of the current database I am connected to.



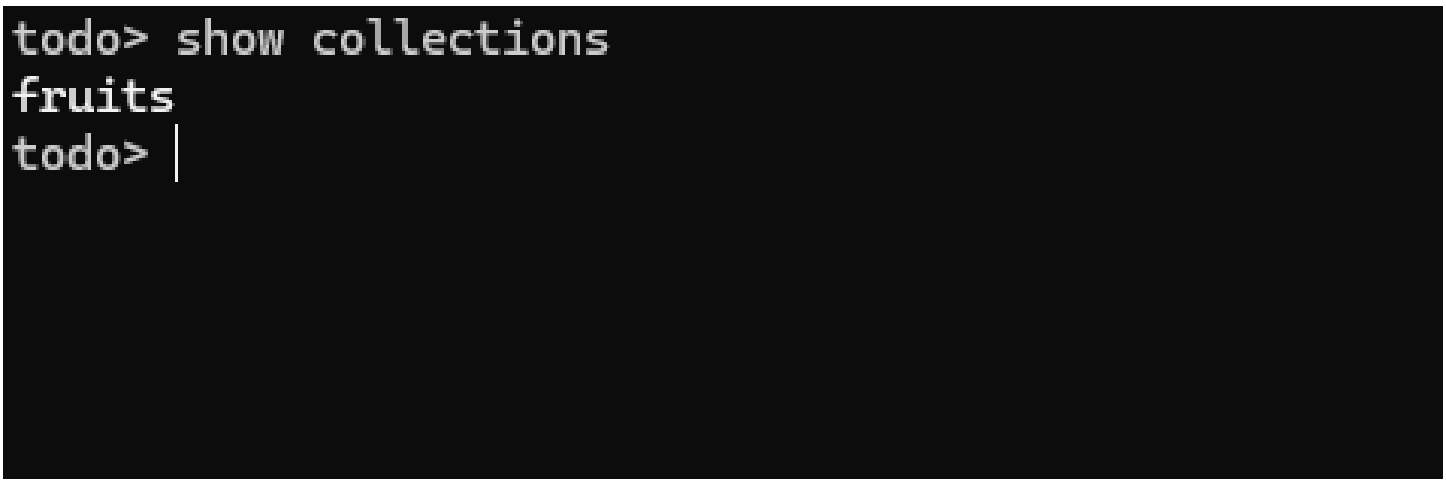
4.**use <myDatabase>**— Used this command to switch to the myDatabase database.



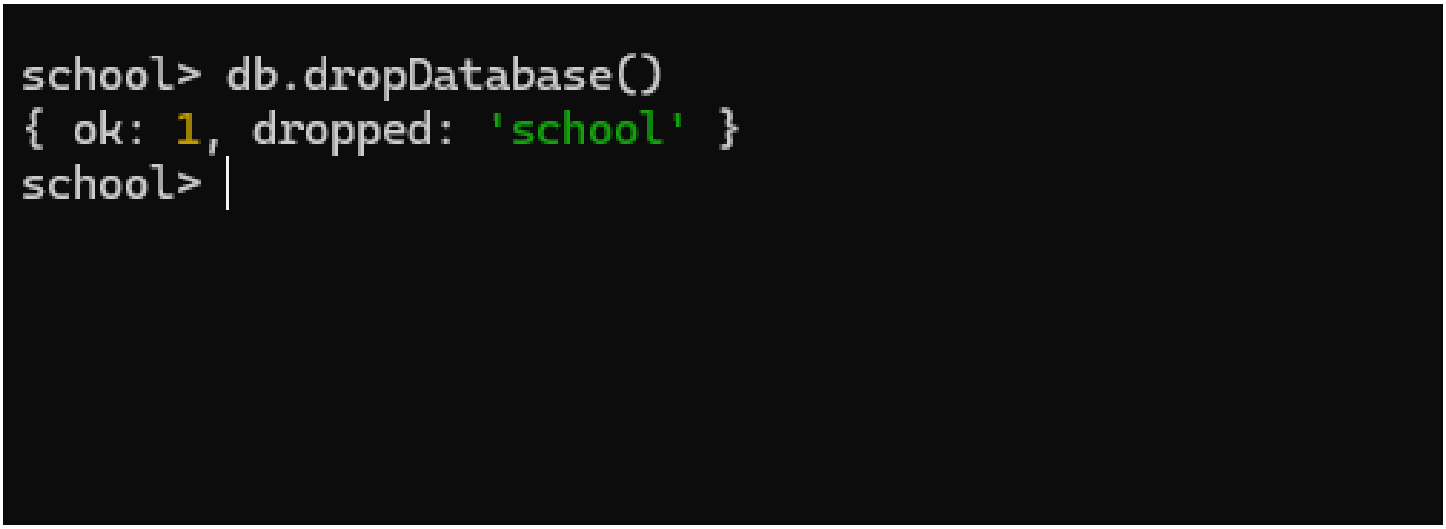
5. **cls** — Used this command to clear the terminal screen and remove the previous commands and output.



6. **show collections** — Used this command to display all the collections available in the current database.



7. **db.dropDatabase()** — Used this command to delete the current database from my MongoDB instance.



8. **exit** — Used this command to exit the current mongosh session and return to the normal terminal.

```
school> exit

C:\Users\saisk>
```

9. **db.users.insertOne({ name: "Kyle" })** — Used this command to add a new document with the name Kyle to the users collection.

```
practice> db.users.insertOne([{name:'kyle'}])
{
  acknowledged: true,
  insertedId: ObjectId('6ac0a0d5bc1a43f91a8dcec7')
}
practice>
```

10. **db.users.insertMany([{ age: 26 }, { age: 20 }])** — Used this command to add two new documents with ages 26 and 20 to the users collection.

```
practice> db.users.insertMany([{age:20},{age:30}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac0a1cabc1a43f91a8dcec8'),
    '1': ObjectId('6ac0a1cabc1a43f91a8dcec9')
  }
}
practice>
```

11. **db.users.find()** — Used this command to retrieve and display all documents from the users collection.

```
practice> db.users.find()
[
  { '0': { name: 'kyle' }, _id: ObjectId('6ac0a0d5bc1a43f91a8dcec7') },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 20 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 }
]
practice>
```

12. **db.users.find({ name: "Kyle" })** — Used this command to retrieve all users whose name is Kyle from the users collection.

```
practice> db.users.find({name:"kyle"})
[ { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' } ]
practice>
```

13 . db.users.find({ name: "Kyle" }, { name: 1, age: 1, _id: 1 }) — Used this command to find users named Kyle and return only their name, age, and _id. **db.users.find({}, { age: 0 })** — Used this command to get all users while excluding the age field.

```
practice> db.users.find({name:"kyle"})
[ { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' } ]
practice> db.users.find({name:"kyle"}, {name:1,age:1,_id:1})
[ { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' } ]
practice> db.users.find({},{age:0})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8') },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9') },
  { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' }
]
practice>
```

14. **db.users.findOne({ name: "Kyle" })** — Used this command to retrieve the first user with the name Kyle.

```
practice> db.users.findOne({name:"kyle"})
{ _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' }
practice>
```

15. **b.users.countDocuments({ name: "Kyle" })** — Used this command to count the number of users with the name Kyle.

```
practice> db.users.countDocuments({name:"kyle"})
1
practice>
```

16. **db.users.updateOne({ age: 20 }, { $set: { age: 21 } })** — Used this command to update the first user with an age of 20 to 21.

```
practice>  db.users.updateOne({age:20},{$set:{age:21}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
practice>
```

17. **db.users.updateMany({ age: 12 }, { $inc: { age: 3 } })** — Used this command to add 3 to the age of all users whose age is 12.

```
practice>  db.users.updateMany({age:12},{$inc:{age:3}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
practice>
```

18. **db.users.replaceOne({ age: 12 }, { age: 13 })** — Used this command to replace the first user with an age of 12 with a new object containing only the age 13.

```
practice> db.users.replaceOne({age:12},{age:13})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
practice>
```

19. **db.users.deleteOne({ age: 20 })** — Used this command to delete the first user with an age of 20.

```
practice> db.users.deleteOne({age:20})
{ acknowledged: true, deletedCount: 0 }
practice>
```

20. **db.users.deleteMany({ age: 12 })** — Used this command to delete all users with an age of 12.

```
practice> db.users.deleteOne({age:20})
{ acknowledged: true, deletedCount: 0 }
practice> db.users.deleteMany({age:12})
{ acknowledged: true, deletedCount: 0 }
practice>
```

21. **db.users.find({ name: { $eq: "Kyle" } })** — Used this command to get all users whose name is exactly Kyle.

```
practice> db.users.find({name:{$eq:"kyle"}})
[ { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' } ]
practice>
```

22. **db.users.find({ name: { $ne: "Kyle" } })** — Used this command to get all users whose name is not Kyle.

```
practice> db.users.find({name:{$ne:"kyle"}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 }
]
practice>
```

23. **db.users.find({ age: { $gt: 12 } })** — Used this command to get all users with an age greater than **db.users.find({ age: { $gte: 15 } })** — Used this command to get all users with an age greater than or equal to 15.

```
practice> db.users.find({age:{$gt:12}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced2'), name: 'Sam', age: 35 }
]
practice>
```

```
practice>  db.users.find({age:{$gte:12}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecc'), name: 'Vijay', age: 12 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced2'), name: 'Sam', age: 35 }
]
practice>
```

24. **db.users.find({ age: { $lt: 12 } })** — Used this command to get all users with an age less than **db.users.find({ age: { $lte: 15 } })** — Used this command to get all users with an age less than or equal to 15.

```
practice> db.users.find({age:{$lt:35}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecb'), name: 'Arun', age: 10 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecc'), name: 'Vijay', age: 12 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 }
]
practice> db.users.find({age:{$lte:35}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecb'), name: 'Arun', age: 10 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecc'), name: 'Vijay', age: 12 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced2'), name: 'Sam', age: 35 }
]
practice>
```

25. **db.users.find({ name: { $in: ["Kyle", "Mike"] } })** — Used this command to get all users whose name is either Kyle or Mike.

```
practice> db.users.find({name:{$in:["kyle","David"]}})
[
  { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 }
]
practice>
```

26. **db.users.find({ name: { $nin: ["Kyle", "Mike"] } })** — Used this command to get all users whose name is neither Kyle nor Mike.

```
practice> db.users.find({name:{$nin:["kyle","David"]}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecb'), name: 'Arun', age: 10 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecc'), name: 'Vijay', age: 12 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced2'), name: 'Sam', age: 35 }
]
practice>
```

27. **db.users.find({ $and: [{ age: 12 }, { name: "Kyle" }] })** — Used this command to get all users who have an age of 12 and the name Kyle

```
practice> db.users.find({$and:[{age:25},{name:"Kyle"}]})
[
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 }
]
practice>
```

28. **db.users.find({ $or: [{ age: 12 }, { name: "Kyle" }] })** — Used this command to get all users with the name Kyle or an age of 12.

```
practice> db.users.find({$or:[{name:"Kyle"},{age:25}]})
[
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 }
]
practice>
```

29. **db.users.find({ name: { $not: { $eq: "Kyle" } } })** — Used this command to get all users whose name is not Kyle.

```
practice>  db.users.find({name:{$not:{$eq:"Kyle"}}})
[
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'), age: 21 },
  { _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'), age: 30 },
  { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecb'), name: 'Arun', age: 10 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecc'), name: 'Vijay', age: 12 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced2'), name: 'Sam', age: 35 }
]
practice>
```

30. **db.users.find({ name: { $exists: true } })** — Used this command to get all users that have a name field.

```
practice> db.users.find({name:{$exists:true}})
[
  { _id: ObjectId('6ac0a32fbc1a43f91a8dceca'), name: 'kyle' },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecb'), name: 'Arun', age: 10 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecc'), name: 'Vijay', age: 12 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecd'), name: 'Karthik', age: 15 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcece'), name: 'Rahul', age: 18 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dcecf'), name: 'David', age: 20 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced0'), name: 'Kyle', age: 25 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced1'), name: 'John', age: 30 },
  { _id: ObjectId('6ac0b178bc1a43f91a8dced2'), name: 'Sam', age: 35 }
]
practice>
```

31. **db.users.find({ $expr: { $gt: ["$balance", "$debt"] } })** — Used this command to get all users whose balance is greater than their debt.

```
practice> db.users.find({$expr:{$gt:["$balance","$debt"]}})
[
  {
    _id: ObjectId('6ac0b834bc1a43f91a8dced3'),
    name: 'Arun',
    balance: 5000,
    debt: 2000
  },
  {
    _id: ObjectId('6ac0b834bc1a43f91a8dced5'),
    name: 'Karthik',
    balance: 7000,
    debt: 4000
  },
  {
    _id: ObjectId('6ac0b834bc1a43f91a8dced8'),
    name: 'Mike',
    balance: 8000,
    debt: 3000
  }
]
practice>
```

32. **db.users.updateOne({ age: 12 }, { $set: { name: "Hi" } })** — Used this command to update the name of the first user with an age of 12 to Hi.

```
practice> db.users.updateOne({age:12},{$set:{name:"hi"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
practice>
```

33. **db.users.updateOne({ age: 12 }, { $inc: { age: 2 } })** — Used this command to add 2 to the age of the first user with an age of 12.

```
practice> db.users.updateOne({age:12},{$inc:{age:2}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
practice>
```

34. **db.users.updateMany({}, { $rename: { age: "years" } })** — Used this command to rename the age field to years for all users.

```
practice> db.users.updateMany({},{$rename:{age:"years"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 17,
  modifiedCount: 10,
  upsertedCount: 0
}
```

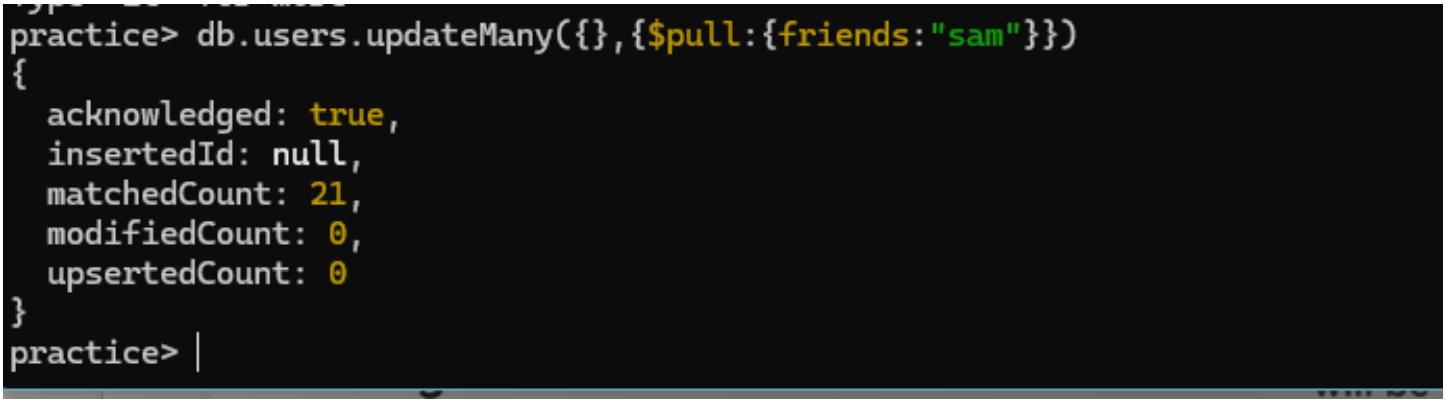
35. **db.users.updateOne({ age: 12 }, { $unset: { age: "" } })** — Used this command to remove the age field from the first user with an age of 12.

```
practice> db.users.updateOne({age:12},{$unset:{age:""}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
practice>
```

36. **db.users.updateMany({}, { $push: { friends: "John" } })** — Used this command to add John to the friends array for all users.

```
practice> db.users.updateMany({},{$push:{friends:"sai"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 21,
  modifiedCount: 21,
  upsertedCount: 0
}
practice> db.users.find()
[
  {
    _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'),
    years: 21,
    friends: [ 'sai' ]
  },
  {
```

37. **db.users.updateMany({}, { $pull: { friends: "Mike" } })** — Used this command to remove Mike from the friends array of all users.



38. **db.users.find().sort({ name: 1, age: -1 })** — Used this command to get all users sorted by name in alphabetical order and then by age in descending order when names are the same.

```
practice> db.users.find().sort({name:1},{age:-1})
[
  {
    _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'),
    years: 21,
    friends: [ 'sai' ]
  },
  {
    _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'),
    years: 30,
    friends: [ 'sai' ]
  },
  {
    _id: ObjectId('6ac0b178bc1a43f91a8dcecb'),
```

39. **db.users.find().limit(2)** — Used this command to return only the first 2 users from the collection.

```
practice> db.users.find().limit(2)
[
  {
    _id: ObjectId('6ac0a1cabc1a43f91a8dcec8'),
    years: 21,
    friends: [ 'sai' ]
  },
  {
    _id: ObjectId('6ac0a1cabc1a43f91a8dcec9'),
    years: 30,
    friends: [ 'sai' ]
  }
```

40. **db.users.find().skip(4)** — Used this command to skip the first 4 users and return the remaining users.

```
practice> db.users.find().skip(4)
[
  {
    _id: ObjectId('6ac0b178bc1a43f91a8dcecc'),
    name: 'hi',
    years: 14,
    friends: [ 'sai' ]
  },
  {
    _id: ObjectId('6ac0b178bc1a43f91a8dcecd'),
    name: 'Karthik',
    years: 15,
    friends: [ 'sai' ]
  },
  {
    _id: ObjectId('6ac0b178bc1a43f91a8dcece'),
    name: 'Rahul',
```